PROVA A1 - ÁLGEBRA LINEAR NUMÉRICA

8 de Abril de 2021

1) $A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$

a) $\xrightarrow[L_4 - L_1]{\substack{L_2 - L_1 \\ L_3 - L_1}} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix} \xrightarrow[L_4 - L_2]{L_3 - L_2} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix}$

$\xrightarrow{L_4 - L_3} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$ 1,0

Assim: $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ e $U = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$

b) $A = LU \Rightarrow \det A = \det L \cdot \det U = a(b-a)(c-b)(d-c)$

Logo: $\boxed{A \text{ é invertível} \leftrightarrow a \neq 0, b \neq a, c \neq b \text{ e } d \neq c}$ 0,5

c) $a=1, b=2, c=3, d=4 \Rightarrow U = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$Ax = v \Rightarrow LUx = v$

$Ux = y \Rightarrow Ly = v \Rightarrow y = \begin{bmatrix} v_1 \\ v_2 - v_1 \\ v_3 - v_2 \\ v_4 - v_3 \end{bmatrix}$ 0,5

$Ux = y \Rightarrow x = \begin{bmatrix} 2v_1 + v_2 \\ 2v_2 - v_1 - v_3 \\ 2v_3 - v_2 - v_4 \\ v_4 - v_3 \end{bmatrix}$

2) $Q_{n\times n}$ ortogonal $\Rightarrow \|Qx\|_2 = \|x\|_2, \ \forall x \in \mathbb{R}^n$.

Assim: $\|Q\|_2 = \max\limits_{\|x\|_2=1} (\|Qx\|_2) = \max\limits_{\|x\|_2=1} (\|x\|_2) = \boxed{1}$ 0,5

$Q_{n\times n}$ ortogonal $\Rightarrow \|q_i\|_2 = 1, \ \forall i=1..n$

Assim: $\|Q\|_F = \sqrt{\sum\limits_{i=1}^{n} \|q_i\|_2^2} = \boxed{\sqrt{n}}$ 0,5

3) a) Sabemos que A^TA é simétrica semidefinida positiva e, portanto, o valor máximo da forma quadrática $x^T(A^TA)x$ com $\|x\|_2=1$ é o raio espectral de A^TA.

Assim: $\|A\|_2 = \max\limits_{\|x\|_2=1} (\|Ax\|_2) = \max\limits_{\|x\|_2=1} \left(\sqrt{(Ax)\cdot(Ax)}\right)$

$= \max\limits_{\|x\|_2=1} \left(\sqrt{x^T(A^TA)x}\right)$

$= \boxed{\sqrt{\rho(A^TA)}}$ 1,0

b) Sendo A simétrica:

$\|A\|_2 = \sqrt{\rho(A^TA)} = \sqrt{\rho(AA)} = \sqrt{\rho(A^2)} = \sqrt{[\rho(A)]^2}$

$= \boxed{\rho(A)}$ 0,5

c) A simétrica definida positiva $\Rightarrow \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$

$\Rightarrow \rho(A) = \lambda_1$ e $\rho(A^{-1}) = \lambda_n^{-1}$.

Logo: $\text{cond}_2(A) = \|A\|_2 \|A^{-1}\|_2$

$= \boxed{\dfrac{\lambda_1}{\lambda_n}}$ 0,5

4) $A_{m \times m}$ invertível

a) Tem-se: $\det(AB - \lambda I) = \det[A^{-1}(AB - \lambda I)A]$

$$= \det(A^{-1}ABA - A^{-1}\lambda I A)$$

$$= \det(BA - \lambda I)$$

1,0

Portanto, AB e BA têm os mesmos polinômios característicos e, assim, têm os mesmos autovalores.

b) Pelo item a da questão 3, tem-se que:

$\|A\|_2 = \sqrt{\rho(A^TA)}$ e $\|A^T\|_2 = \sqrt{\rho(AA^T)}$.

Entretanto, pelo item anterior, $\rho(A^TA) = \rho(AA^T)$.

Logo: $\boxed{\|A\|_2 = \|A^T\|_2}$

0,5

c) Tem-se:

$$\mathrm{cond}_2(A^T) = \|A^T\|_2 \|(A^T)^{-1}\|_2$$

$$= \|A^T\|_2 \|(A^{-1})^T\|_2$$

$$= \|A\|_2 \|A^{-1}\|_2$$

$$= \mathrm{cond}_2(A)$$

0,5

5) $A = \begin{bmatrix} \alpha & -1 & 0 & -1 \\ -1 & \alpha & -1 & 0 \\ 0 & -1 & \alpha & -1 \\ -1 & 0 & -1 & \alpha \end{bmatrix}$

a) Autovalores: $\begin{vmatrix} \alpha-\lambda & -1 & 0 & -1 \\ -1 & \alpha-\lambda & -1 & 0 \\ 0 & -1 & \alpha-\lambda & -1 \\ -1 & 0 & -1 & \alpha-\lambda \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} \alpha-\lambda-2 & -1 & 0 & -1 \\ \alpha-\lambda-2 & \alpha-\lambda & -1 & 0 \\ \alpha-\lambda-2 & -1 & \alpha-\lambda & -1 \\ \alpha-\lambda-2 & 0 & -1 & \alpha-\lambda \end{vmatrix} = 0$

substituindo a 1ª coluna pela soma dela com as demais.

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Substituindo as 2ª, 3ª e 4ª linhas pelas diferenças entre cada uma delas e a 1ª linha, tem-se:

$$\begin{vmatrix} \alpha-\lambda-2 & -1 & 0 & -1 \\ 0 & \alpha-\lambda+1 & -1 & 1 \\ 0 & 0 & \alpha-\lambda & 0 \\ 0 & 1 & -1 & \alpha-\lambda+1 \end{vmatrix} = 0 \Rightarrow (\alpha-\lambda-2)\begin{vmatrix} \alpha-\lambda+1 & -1 & 1 \\ 0 & \alpha-\lambda & 0 \\ 1 & -1 & \alpha-\lambda+1 \end{vmatrix} = 0$$

$$\Rightarrow (\alpha-\lambda-2)\left[(\alpha-\lambda+1)^2(\alpha-\lambda) - (\alpha-\lambda)\right] = 0$$

1,0

$$\Rightarrow (\alpha-\lambda-2)(\alpha-\lambda)\left[(\alpha-\lambda+1)^2-1\right] = 0$$

$\boxed{\lambda = \alpha - 2}$, $\boxed{\lambda = \alpha}$

$\alpha-\lambda+1 = 1 \Rightarrow \boxed{\lambda = \alpha}$

$\alpha-\lambda+1 = -1 \Rightarrow \boxed{\lambda = \alpha + 2}$

b) Deve-se ter: $\begin{cases} \alpha - 2 > 0 \Rightarrow \boxed{\alpha > 2} \\ \boxed{\alpha > 0} \\ \alpha + 2 > 0 \Rightarrow \boxed{\alpha > -2} \end{cases} \Rightarrow \boxed{\alpha > 2}$

0,5

c) $\underline{A}$ é singular $\Leftrightarrow$ $\lambda = 0$ é autovalor de A

$\Leftrightarrow \boxed{\alpha = -2 \text{ ou } \alpha = 0 \text{ ou } \alpha = 2}$

0,5

d) Tem-se: $M_J = -D^{-1}(L+U) = -\begin{bmatrix} \frac{1}{\alpha} & 0 & 0 & 0 \\ 0 & \frac{1}{\alpha} & 0 & 0 \\ 0 & 0 & \frac{1}{\alpha} & 0 \\ 0 & 0 & 0 & \frac{1}{\alpha} \end{bmatrix}\begin{bmatrix} 0 & -1 & 0 & -1 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ -1 & 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{\alpha} & 0 & \frac{1}{\alpha} \\ \frac{1}{\alpha} & 0 & \frac{1}{\alpha} & 0 \\ 0 & \frac{1}{\alpha} & 0 & \frac{1}{\alpha} \\ \frac{1}{\alpha} & 0 & \frac{1}{\alpha} & 0 \end{bmatrix}$

$\boxed{\text{O método converge} \Leftrightarrow \rho(M_J) < 1}$

Assim:

$$\begin{vmatrix} -\lambda & \frac{1}{\alpha} & 0 & \frac{1}{\alpha} \\ \frac{1}{\alpha} & -\lambda & \frac{1}{\alpha} & 0 \\ 0 & \frac{1}{\alpha} & -\lambda & \frac{1}{\alpha} \\ \frac{1}{\alpha} & 0 & \frac{1}{\alpha} & -\lambda \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -\lambda+\frac{2}{\alpha} & \frac{1}{\alpha} & 0 & \frac{1}{\alpha} \\ -\lambda+\frac{2}{\alpha} & -\lambda & \frac{1}{\alpha} & 0 \\ -\lambda+\frac{2}{\alpha} & \frac{1}{\alpha} & -\lambda & \frac{1}{\alpha} \\ -\lambda+\frac{2}{\alpha} & 0 & \frac{1}{\alpha} & -\lambda \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -\lambda+\frac{2}{\alpha} & \frac{1}{\alpha} & 0 & \frac{1}{\alpha} \\ 0 & -\lambda-\frac{1}{\alpha} & \frac{1}{\alpha} & -\frac{1}{\alpha} \\ 0 & 0 & -\lambda & 0 \\ 0 & -\frac{1}{\alpha} & \frac{1}{\alpha} & -\lambda-\frac{1}{\alpha} \end{vmatrix} = 0$$

$$\Rightarrow \left(-\lambda+\frac{2}{\alpha}\right)\left[\left(-\lambda-\frac{1}{\alpha}\right)^2(-\lambda) - (-\lambda)\frac{1}{\alpha^2}\right] = 0 \Rightarrow \left(-\lambda+\frac{2}{\alpha}\right)(-\lambda)\left[\left(-\lambda-\frac{1}{\alpha}\right)^2 - \frac{1}{\alpha^2}\right] = 0 \Rightarrow$$

1,0

$\Rightarrow \lambda = \frac{2}{\alpha}$, $\lambda = 0$ (dupla) e $\lambda = -\frac{2}{\alpha}$. Logo, deve-se ter $\left|\frac{2}{\alpha}\right| < 1 \Rightarrow \boxed{|\alpha| > 2}$